3.1.4 Energetics: Sheet 1

Enthalpy changes · 12 marks · about 25 minutes

Key information

- **Enthalpy change (ΔH):** heat energy change at constant pressure. Exothermic: ΔH negative; endothermic: ΔH positive.
- Standard conditions: 100 kPa and a stated temperature (usually 298 K), with all substances in their standard states.
- **Standard enthalpy of formation ($\Delta_f H^\ominus$):** enthalpy change when 1 mole of a substance is formed from its elements, all in their standard states.
- **Standard enthalpy of combustion ($\Delta_c H^\ominus$):** enthalpy change when 1 mole of a substance is burned completely in oxygen, all in their standard states.
- $\Delta_f H$ equations must make exactly 1 mole of product, so fractions such as $\frac{1}{2}O_2$ are often needed.

Section A: Self-quiz

01 Which change occurs when water is vaporised?

[1 mark]

A An exothermic change occurs.
B Covalent bonds are broken.
C Intermolecular forces are overcome.
D The total energy of the molecules decreases.

02 Which equation represents the reaction that has a standard enthalpy change equal to the standard enthalpy of formation for barium chloride?

[1 mark]

A $Ba(g) + Cl_2(g) \rightarrow BaCl_2(s)$
B $Ba^{2+}(g) + 2\ Cl^-(g) \rightarrow BaCl_2(s)$
C $Ba(s) + Cl_2(g) \rightarrow BaCl_2(s)$
D $Ba^{2+}(s) + 2\ Cl^-(g) \rightarrow BaCl_2(s)$

03 Which reaction has an enthalpy change equal to the standard enthalpy of formation of potassium oxide?

[1 mark]

A $4\ K(s) + O_2(g) \rightarrow 2\ K_2O(s)$
B $2\ K(s) + O(g) \rightarrow K_2O(s)$
C $2\ K^+(g) + O^{2-}(g) \rightarrow K_2O(s)$
D $2\ K(s) + \frac{1}{2}O_2(g) \rightarrow K_2O(s)$

04 Which reaction has an enthalpy change equal to the standard enthalpy of formation of lithium fluoride?

[1 mark]

A $Li(g) + \frac{1}{2}F_2(g) \rightarrow LiF(s)$
B $Li^+(g) + F^-(g) \rightarrow LiF(s)$
C $Li^+(aq) + F^-(aq) \rightarrow LiF(s)$
D $Li(s) + \frac{1}{2}F_2(g) \rightarrow LiF(s)$

05 State the meaning of the term enthalpy change as applied to a chemical reaction.

[1 mark]

Section B: Exam questions

06 Define standard enthalpy of formation.

[2 marks]

07 State the meaning of the term standard enthalpy of combustion.

[2 marks]

08 Write an equation, including state symbols, for the reaction with an enthalpy change equal to the enthalpy of formation for iron(III) oxide.

[1 mark]

09 Silver nitrate(V) is formed when silver nitrate(III) undergoes thermal decomposition.

$2\ AgNO_2(s) \rightarrow Ag(s) + AgNO_3(s) + NO(g)$

Suggest why the enthalpy change for the thermal decomposition of solid silver nitrate(III) is difficult to determine experimentally.

[1 mark]

10 Anhydrous magnesium chloride, $MgCl_2$, can absorb water to form the hydrated salt $MgCl_2.4H_2O$

$MgCl_2(s) + 4H_2O(l) \rightarrow MgCl_2.4H_2O(s)$

Suggest one reason why the enthalpy change for this reaction cannot be determined directly by calorimetry.

[1 mark]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**C**	Source: 7404/1 Jun 2019 Q13	1
02	**C**	Source: 7404/1 Jun 2019 Q14	1
03	**D**	Source: 7404/1 Jun 2023 Q15	1
04	**D**	Source: 7404/1 Nov 2020 Q12	1
05	Heat energy change at constant pressure	Source: 7404/1 Nov 2020 Q04.1	1
06	M1 The enthalpy change when one mole of a substance is formed from its (constituent) elements M2 All substances in their standard states (at 100 kPa and a stated temperature)	Source: 7404/1 Jun 2024 Q08.1 Allow heat energy change	2
07	M1 Enthalpy change when one mole of a substance burns completely in oxygen M2 With all substances in their standard states (at stated temperature and 100 kPa)	Source: 7404/1 Nov 2021 Q07.1 Allow heat energy change / allow fully combust	2
08	$2Fe(s) + 1½O_2(g) \rightarrow Fe_2O_3(s)$ ONLY	Source: 7404/1 Jun 2017 Q02.1 Do not allow multiples. State symbols must be shown	1
09	The silver nitrate has to be heated so you cannot measure the temperature change caused by the reaction	Source: 7404/1 Jun 2024 Q08.3 Allow difficult to measure the temperature (change) of solids / $AgNO_3$ may decompose / other oxides of nitrogen would be formed	1
10	Not possible to prevent some dissolving	Source: 7405/3 Jun 2017 Q01.1 Allow it is soluble / dissolves / other hydrates may form / suggestions related to difficulty of measuring T (change) of a solid	1